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Canadian Bioinformatics Workshops



Module 4: Genome Assembly, Polishing, Evaluation and Annotation

Dr. Ido Hatam

CBW High-Throughput Genomics

November 4-6, 2026

Langara.

ʔəx̌ʷməθkʷəy̓əm | at Musqueam

Learning Objectives

- Explain the difference between de novo and reference-based assembly
- Explain how polishing improves assembly accuracy
- Explain how contiguity, correctness and completeness are evaluated
- Read and interpret Flye, QUAST and BUSCO reports
- Distinguish structural from functional annotation and read their outputs (GFF3, eggNOG table)

CANADIAN BIOINFORMATICS HUB

— MAIN EVENT —

COMPARATIVE GENOMICS WORKSHOP

THE CORGI

S288C · THE REFERENCE

Has never once been outside the lab.



TALE OF THE TAPE

ORIGIN
Lab-bred, 1950s

PLOIDY
Haploid (MATα)

GENOME
12.1 Mb · 16 chr

WEAKNESS
flo8: can't form a pack

WEAKNESS
ho: needs a matchmaker

VS

THE WOLF

BAL_1a · WILD ISOLATE (ScRAP)

Doesn't do coordinates.

TALE OF THE TAPE

ORIGIN
Wild, China: the species' homeland

PLOIDY
???

GENOME
??? Mb · ??? chr

SVs VS. THE CORGI
???

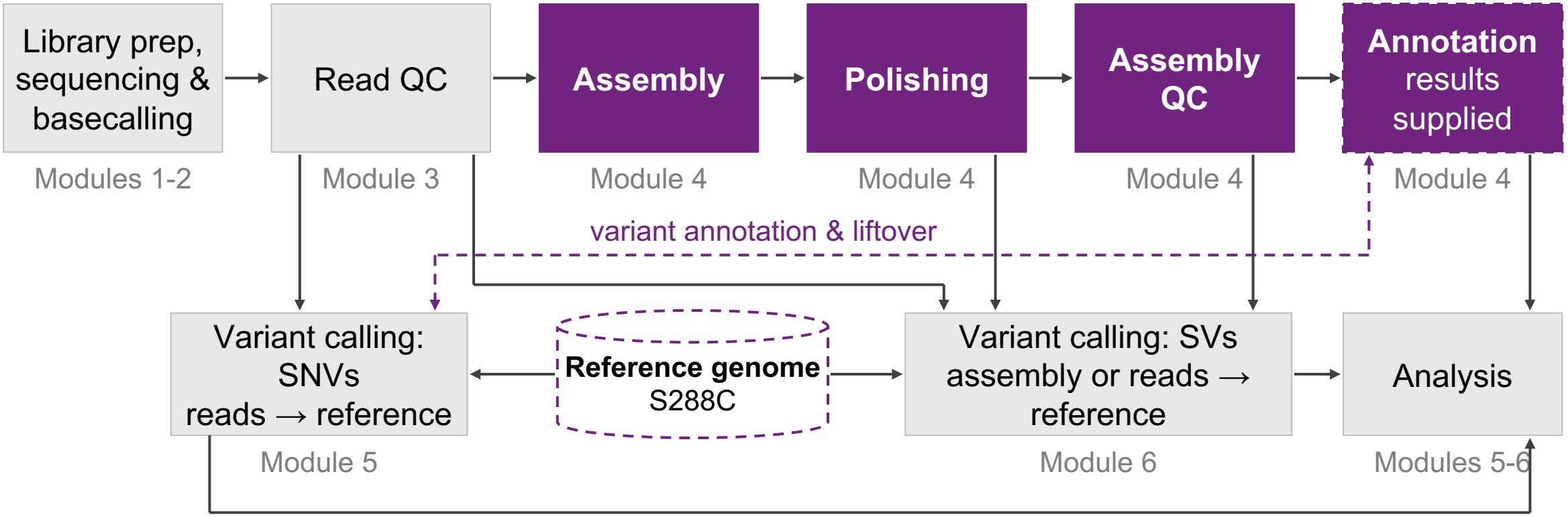
Stats revealed round by round. By you.



FIGHT CARD

ROUND 1	ROUND 2	ROUND 3	ROUND 4	ROUND 5
QC	ASSEMBLY	ANNOTATION	WHOLE-GENOME ALIGNMENT	SVs & VARIANTS

Where we are in the workflow

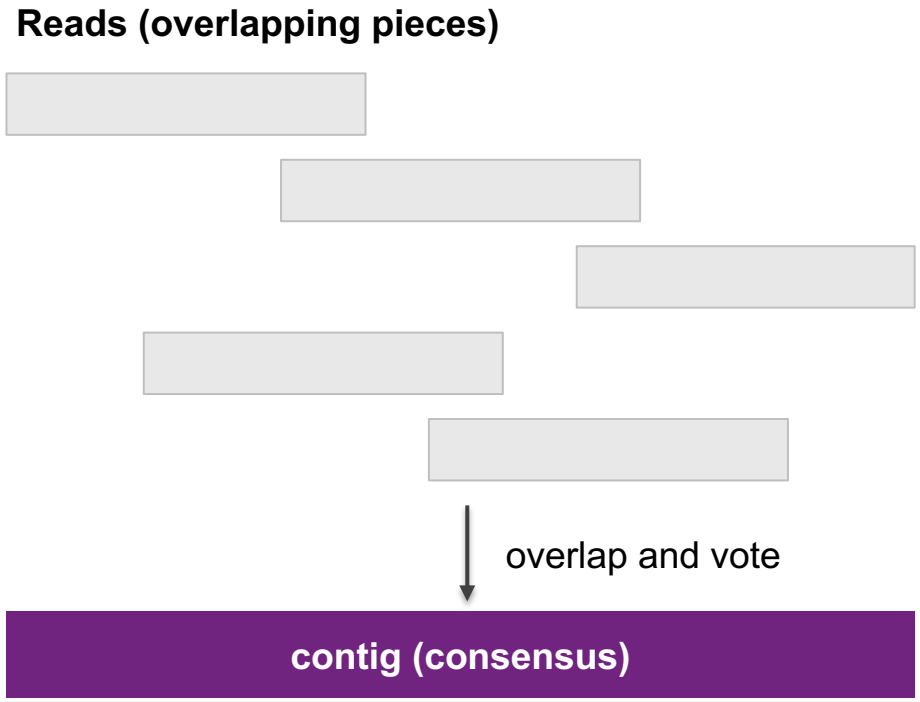


= this module (dashed outline = supplied or reference)

The case: can we rebuild the wolf's genome, one piece per chromosome?

What is genome assembly?

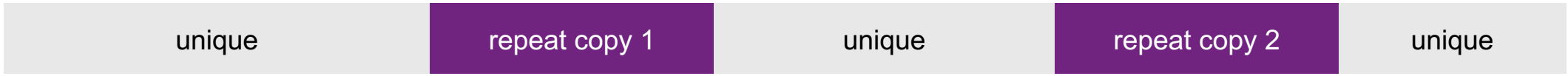
- Sequencing gives millions of pieces, reads.
  - None spans a chromosome
- Assembly: rebuild the genome from reads
- Overlapping reads are joined; the consensus is kept
- The result: contigs, saved in a FASTA file
- Like a jigsaw puzzle



Schematic: reads shortened for the picture

Repeats: the "blue sky" problem

Genome



Short reads



which copy?

Long read



spans the repeat and both flanks

Read **shorter** than the repeat: its origin is ambiguous, so the assembly breaks

Read **longer** than the repeat: unique flanks anchor it, so the pieces connect

Contigs and scaffolds

Contigs (gap-free)



Scaffold (contigs joined by N gaps)



Contig: a gap-free sequence

Scaffold: contigs joined with N gaps (**N** = unknown base)

Goal for yeast: one contig per chromosome. ScRAP assembled the wolf (BAL_1a) as 16 nuclear contigs, 32 telomeres, 0 gaps. How close will we get?

De novo vs reference-guided

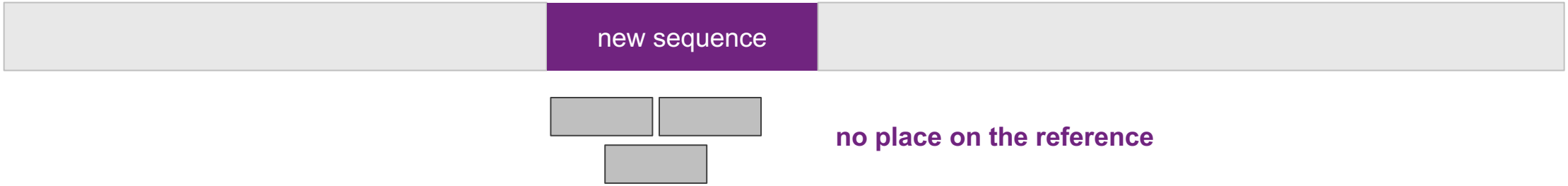
	De novo	Reference-guided
Idea	Compare reads with each other; no guide	Place each read on a reference genome
Strength	Finds new sequence and structural variants	Fast; little memory
Weakness	Heavy on memory and CPU	Reference bias: misses what the reference lacks
Use when	No reference, or a very different sample	A close reference exists

Reference bias

Reference genome



Your sample



Reads from sequence the reference lacks are lost or misplaced

How different are the wolf (BAL_1a) and the corgi (S288C)? Modules 5 and 6 measure it

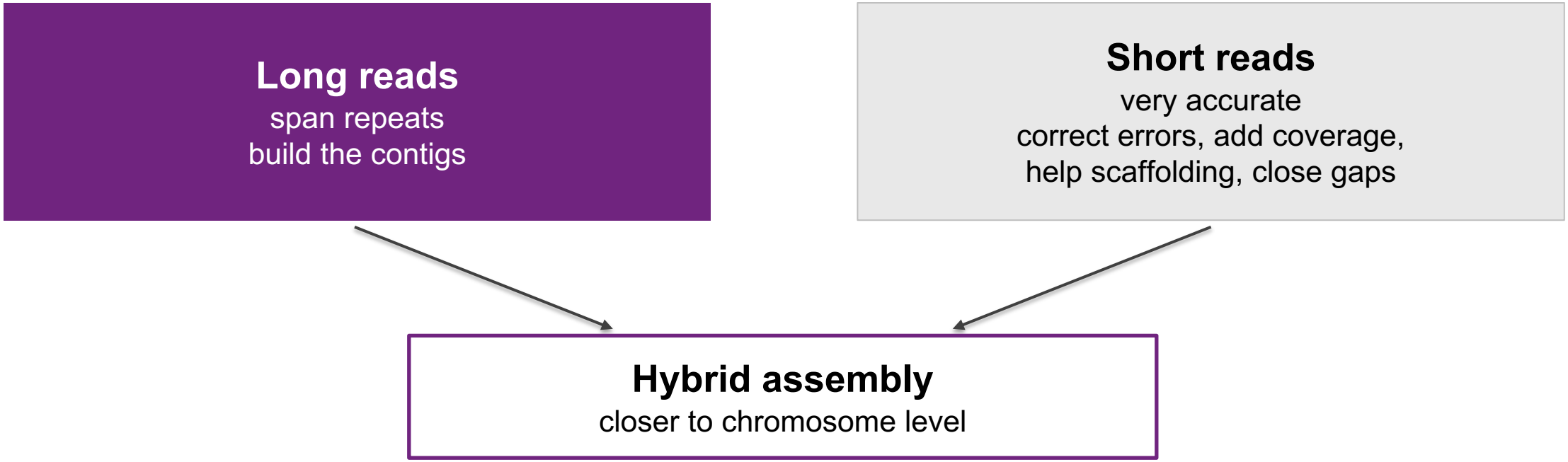
Your turn: which strategy?

- A new yeast species with no genome sequence yet?
- 200 strains of S. cerevisiae, to find SNVs (single-base differences) in each?
- One wild isolate where you suspect large rearrangements?
- **Our pipeline:** de novo with Flye first, then the reference for evaluation and variant calling

Long vs short reads for assembly

	Short reads (Illumina)	Long reads (ONT)
Length	about 150-300 bp	thousands to 100,000+ bp (ours, Lab 3: mean about 19.5 kb)
Accuracy	about 99.9% (Q30) for most bases	Chemistry dependent
Repeats	Longer than the read: assembly breaks	Spanned by one read: contigs connect
Assembler	de Bruijn graph	overlap or repeat graph (Flye)
Needs	High coverage	Polishing

Hybrid assembly: long plus short



Illumina reads of this isolate exist (ENA ERR8603864); we do not use them in this session because of time. Polishing carries the accuracy.

How assemblers solve the puzzle

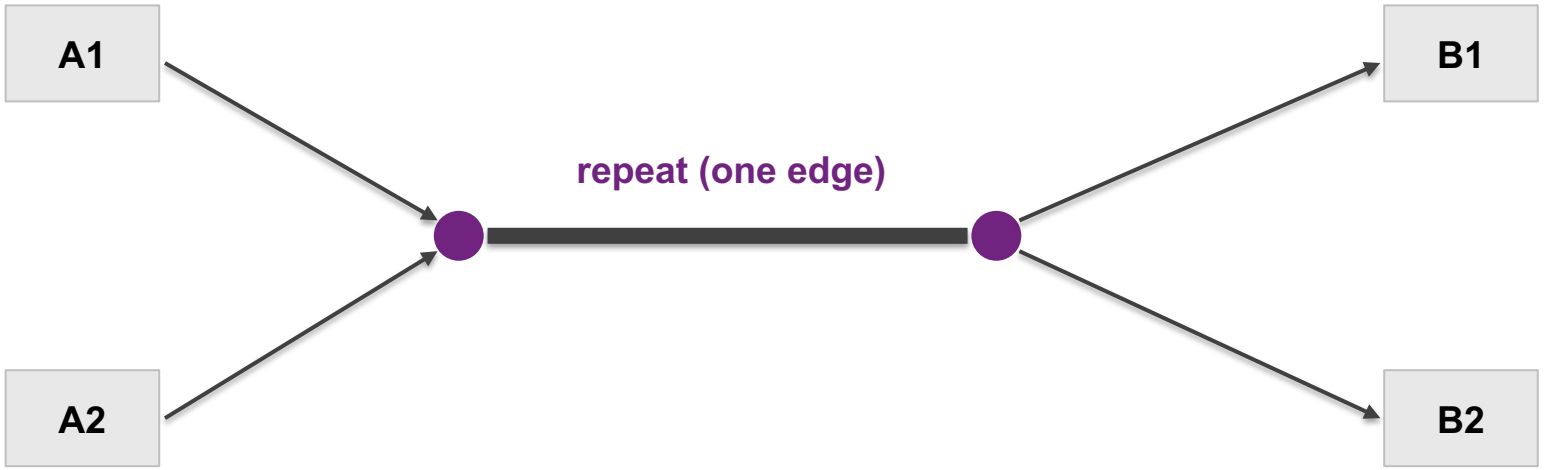
Greedy
merge the best overlap;
stalls on repeats

→

de Bruijn graph
reads cut into k-mers;
standard for short reads

→

Repeat graph (Flye)
long reads; repeats
resolved by spanning reads



Which exit goes with which entrance?

A read spanning A1, the repeat and B1 answers it

Tools you may meet with short reads: assembly

Tool	What it does
SPAdes	De Bruijn graph assembler for short reads (also hybrid)
MaSuRCA	Short-read or hybrid assembly

Not covered in this workshop

Flye: our assembler

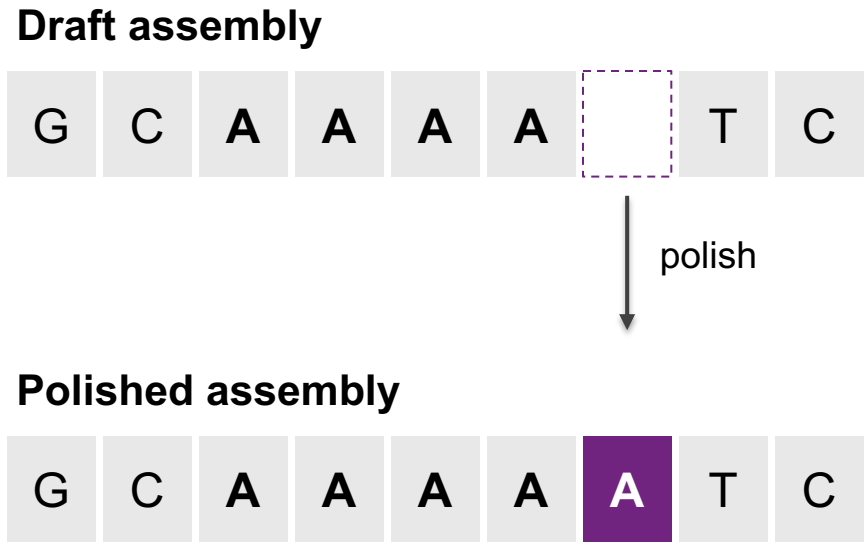
- Built for long, noisy reads; builds a repeat graph
- Read-type flag must match the data
- --nano-raw: older ONT chemistries (up to 20% error)
- --nano-hq: ONT R10 reads, aka Q20 (<3% error)
- Ours: R9.4.1, basecaller unknown; about Q9 (Lab 3) fits --nano-raw

```
flye --nano-raw reads.fastq.gz
  --genome-size 12m
  --iterations 3
  --threads 8
  --out-dir assembly/
```

genome size: our estimate (about 12 Mb)
iterations: Flye's own polishing rounds
runs about 20 minutes
the log reports total length, contigs, N50 and coverage

Why polish?

- The assembler fixes the order of the pieces, but single bases can still be wrong
- Polishing: correct base errors in the assembly using the reads
- Typical ONT error: a lost base in a run of one base (a homopolymer)
- Flye polishes internally first; then medaka polishes its consensus
- Polishing fixes bases, not structure



Illustrative example

Polishing: map and vote

Draft assembly (one wrong base)

A	C	G	T	**T**	A	C	G	A	T

Reads mapped to the draft

G	T	**C**	A	C	G	A
G	T	**C**	A	C	G	A
G	T	**C**	A	C	G	A
G	T	**C**	A	C	G	A
G	T	T	A	C	**A**	A

4 reads say C: vote C

Polished assembly

A	C	G	T	**C**	A	C	G	A	T

1 Map the reads to the draft

2 Stack: look at all bases at each position

3 Vote: the majority base replaces the draft base

One read's random error (here A) is outvoted

medaka: a neural-network polisher

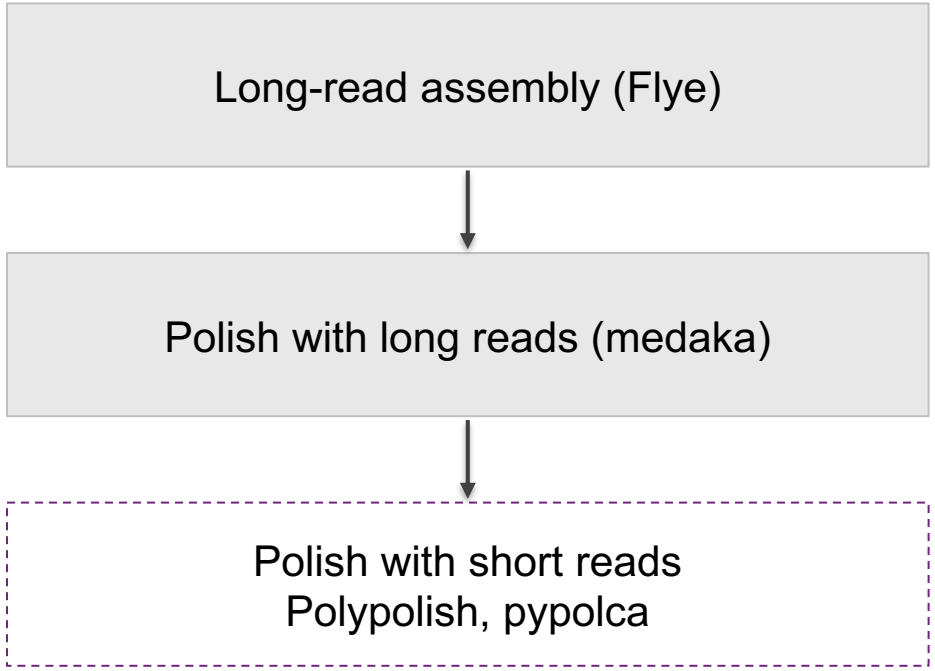
- Made by Oxford Nanopore
- A neural network reads the stack of reads and decides the consensus
- It learned systematic nanopore error patterns, e.g. homopolymers
- Models are chemistry-specific: match flow cell and basecaller

Our model name, decoded

r941	min	sup	g507
R9.4.1 flow cell	MinION device	super-accuracy model	Guppy 5.0.7 (training data)

Hybrid polishing with short reads

- Illumina reads of this isolate (ERR8603864, slide 13): we do not use them in this session because of time
- Very accurate (Q30): map them to the assembly and vote again
- Tools you may meet: Polypolish, pypolca (reimplements POLCA)
- Short reads map poorly inside repeats: less reliable there
- After the workshop: SPAdes vs Flye vs Flye + short-read polishing



dashed = not covered in this workshop

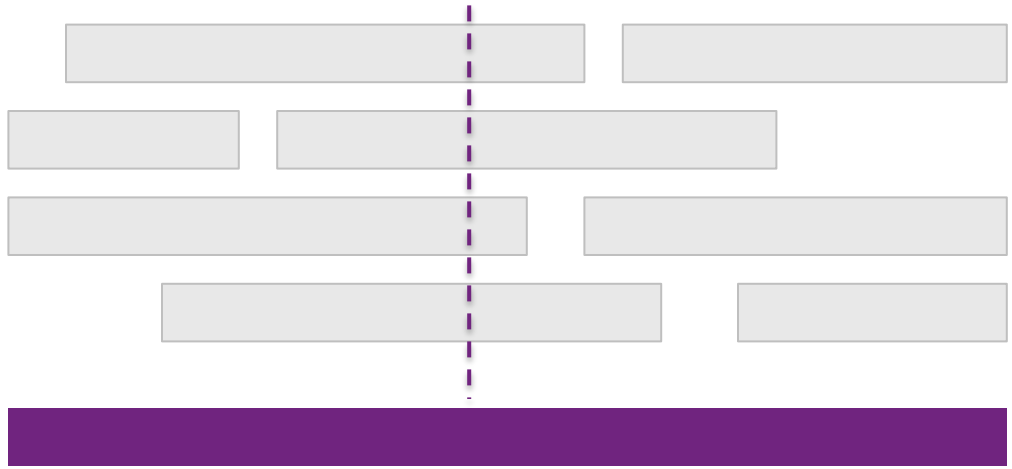
Judging an assembly: three lenses

Contiguity	Correctness	Completeness
How few, how long are the pieces? **contigs, N50, N90, L50** Tool: Flye log, QUAST	Does it agree with the reads and a reference? **misassemblies, mismatches** Tool: QUAST with a reference	Are the expected genes there? **C, S, D, F, M** Tool: BUSCO

Coverage underlies all three: how much evidence the assembly rests on. Never judge on one number

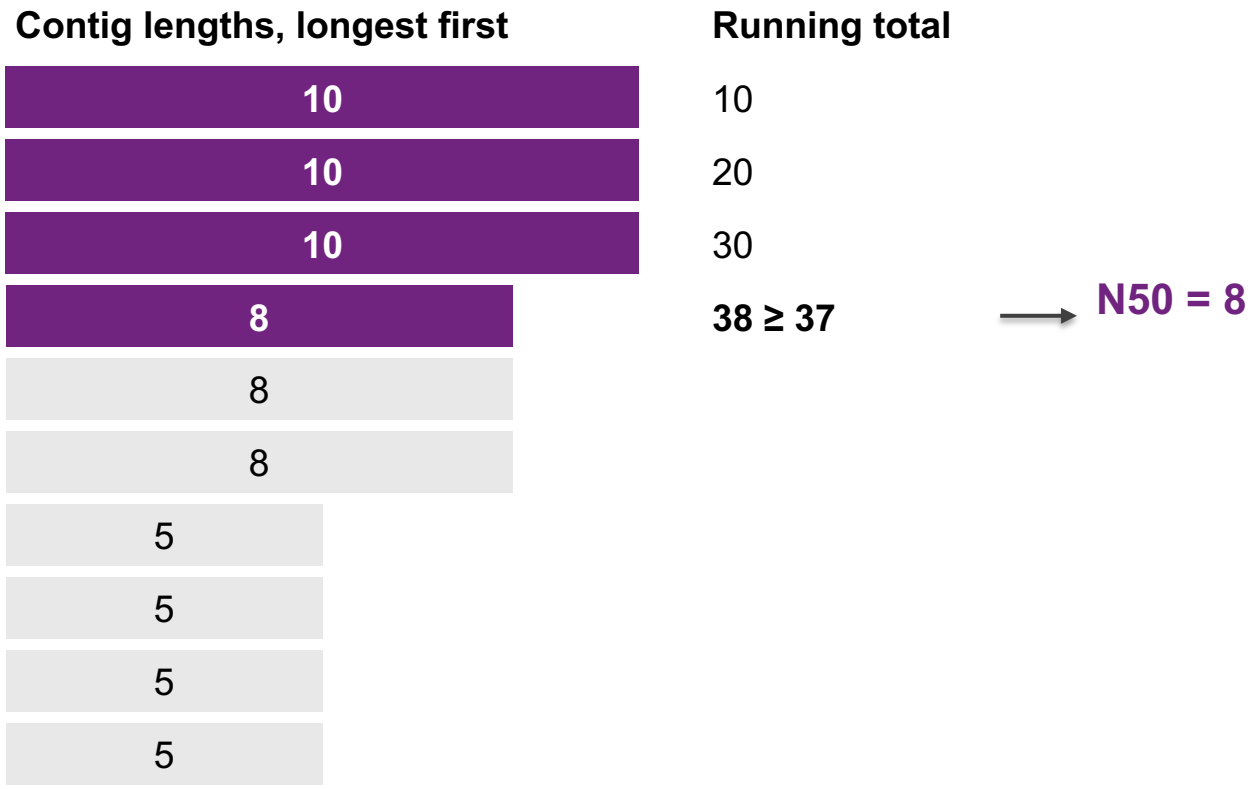
Coverage: how much evidence?

- **Coverage (depth)**: the average number of reads covering each base
- Total sequenced bases ÷ genome size
- Ours (Lab 3): **448 Mb ÷ 12.1 Mb = about 37×**
- Long-read assemblies: aim for 30-50×
- The Flye log reports the mean coverage



Schematic

N50: a worked example



Total = 74; half = 37

Add contigs from the longest until you pass 37

It takes 4 contigs: **L50 = 4**

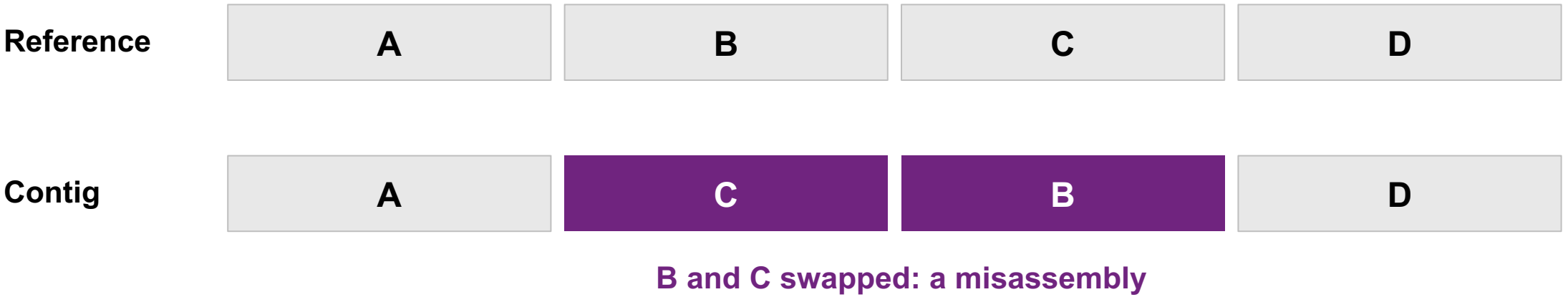
The last one added has length 8: **N50 = 8**

QUAST: the corgi's card

- **QUAST**: Quality Assessment Tool for Genome Assemblies
- No reference needed for the basics
- Here: the S288C reference, our yardstick: 17 pieces
- GC 38%: the yeast value we compare against; 0 Ns: no gaps
- Report as text and HTML

S288C reference	QUAST
# contigs	17
Largest contig	1,531,933
Total length	12,157,105
GC (%)	38.15
N50	924,431
N90	439,888
L50	6
# N's per 100 kbp	0.00

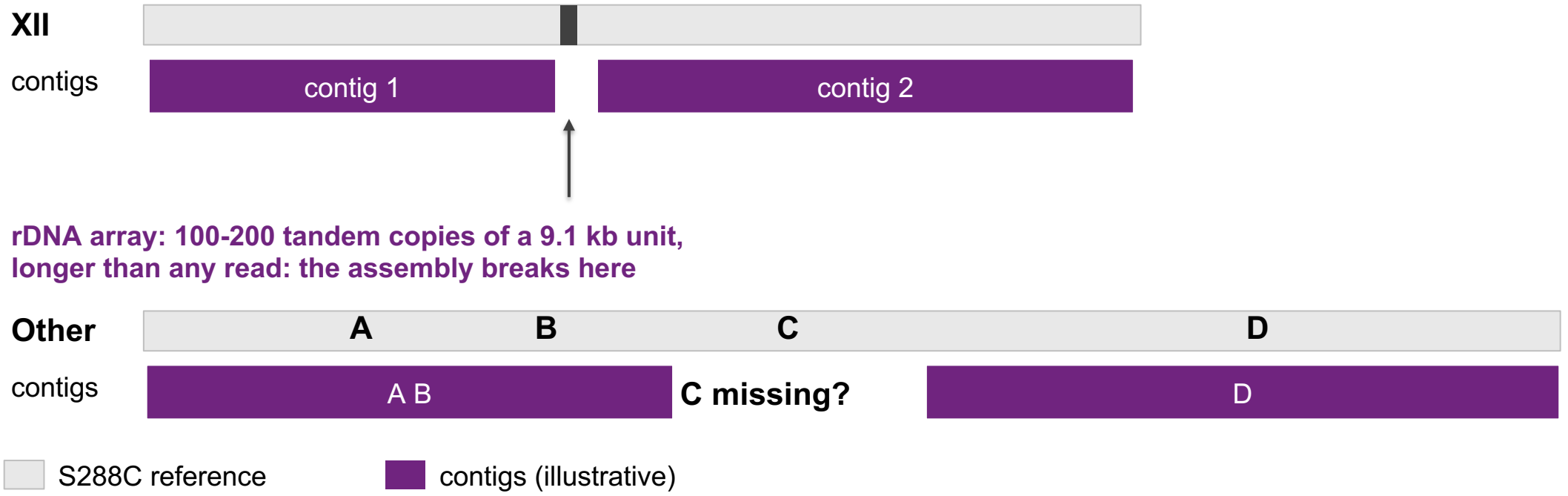
QUAST with a reference



Misassembly: order or orientation of blocks disagrees with the reference

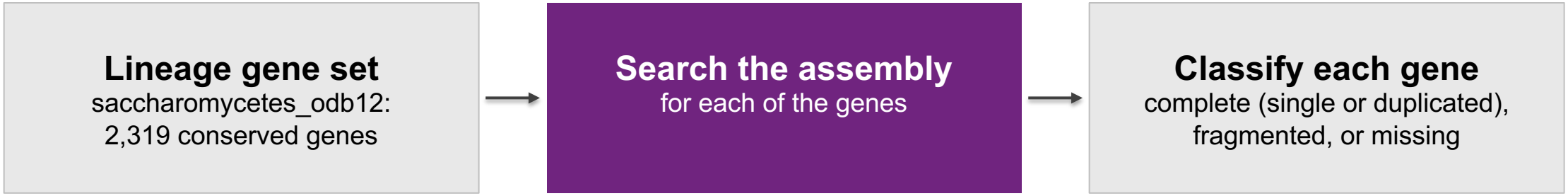
Also reported: mismatches and indels per 100 kbp, and the share of the reference covered

Misassembly or real variant?



- XII: the rDNA break is a contig end (contiguity), an assembly limit, not a misassembly
- A block missing against the reference: assembly error, or a real difference between the wolf and the corgi? QUAST cannot tell

BUSCO: are the genes there?



BUSCO: Benchmarking Universal Single-Copy Orthologs

Ortholog: the same gene in different species, split by speciation

These genes are expected once in nearly every yeast of the lineage

A biological check: does the assembly contain the genes a yeast needs?

Reading a BUSCO result

C:43.7%[S:43.4%,D:0.3%],F:2.6%,M:53.7%,n:2319

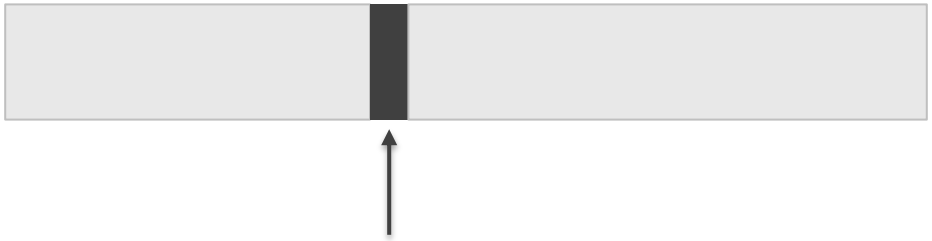
	Meaning	R7.3 chemistry	Genes
C	Complete (C = S + D)	43.7%	1,014
S	Complete, single-copy	43.4%	1,006
D	Complete, duplicated	0.3%	8
F	Fragmented	2.6%	60
M	Missing	53.7%	1,245
n	BUSCO genes searched		2,319

Note: Percentages are of the 2,319 BUSCO genes

What each BUSCO class tells you

- **S**: found once, as expected
- **D**: found more than once: real paralogs, or a region assembled twice
- **F**: only part of the gene: a gap or errors
- **M**: the gene is absent
- **Internal stop codons (E):** a complete gene with a stop inside. Legacy: 70 of 1,014 (6.9%)

A complete gene with an internal stop



stop codon inside the gene

A conserved gene should not have a stop in the middle. Which nanopore errors (Module 3) could cause this?

How many in ours, and why? Lab 4

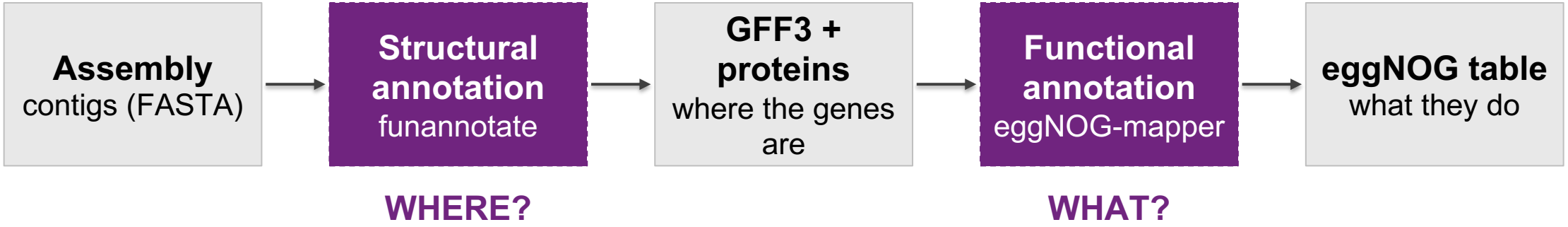
A cautionary example: the legacy R7.3 assembly

Legacy R7.3 (ERR1539006): BUSCO genes



	Legacy R7.3	S288C (yardstick)	Ours: Lab 4
Contigs	431	17	?
Total length (bp)	10,073,637	12,157,105	?
N50 (bp)	28,584	924,431	?
BUSCO complete (C)	43.7%	not run	?
BUSCO missing (M)	53.7%	not run	?

Annotation: where, then what



Example: ADH1 in the corgi's annotation (S288C, Ensembl)

Structural
gene YOL086C, chromosome XV:159,548-160,594, minus strand

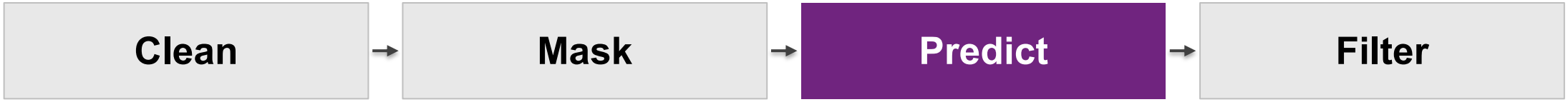
→

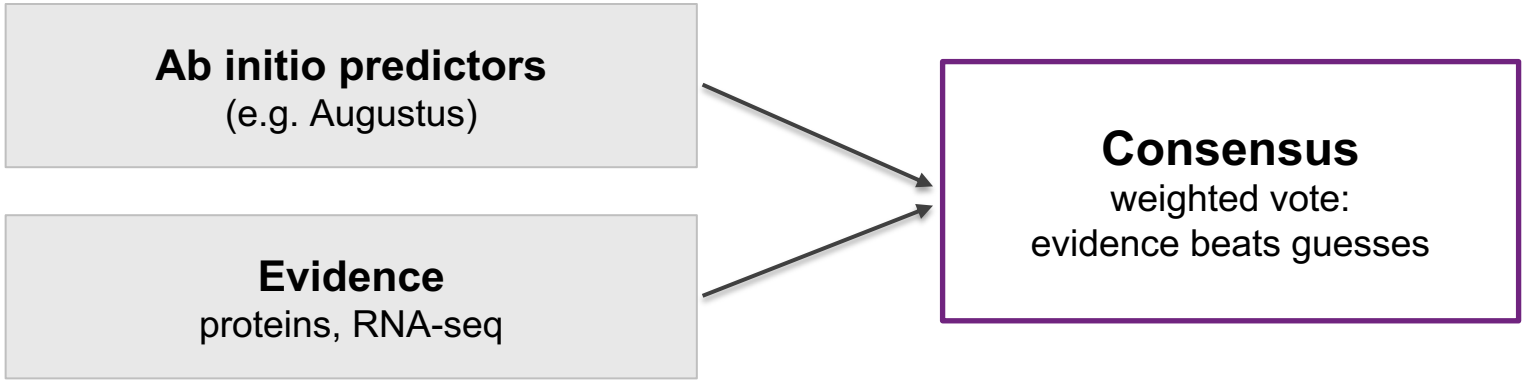
Functional
named ADH1: alcohol dehydrogenase, acetaldehyde to ethanol

Finding genes: two approaches

	Ab initio	Evidence-based
Idea	Statistical models find gene patterns: start and stop codons, splice sites	Align known proteins and RNA-seq reads to the genome
Strength	Fast; needs no extra data	Real support; RNA-seq shows what is transcribed
Weakness	False positives if the model is poorly trained	Needs data; proteins find only known genes; RNA-seq misses unexpressed genes

funannotate: the pipeline





Yeast versus larger eukaryotes

	Yeast	Human (a larger eukaryote)
Genome size	about 12 Mb	about 3,100 Mb
Protein-coding genes	about 6,000	about 20,000
Genes with introns	few (about 5%)	most
Alternative splicing	rare	common
Gene prediction	ab initio works well on accurate sequence	needs RNA-seq evidence

A prediction is a hypothesis: compare it with what you expect for the species

How many gene models in ours, and how many are split into pieces? Lab 4

GFF3: the annotation file

seqid	source	type	start	end	score	strand	phase	attributes
XV	sgd	gene	159548	160594	.	-	.	ID=gene:YOL086C;Name=ADH1
XV	sgd	mRNA	159548	160594	.	-	.	ID=transcript:YOL086C_mRNA;Parent=gene:YOL086C
XV	sgd	exon	159548	160594	.	-	.	Parent=transcript:YOL086C_mRNA
XV	sgd	CDS	159548	160594	.	-	0	ID=CDS:YOL086C;Parent=transcript:YOL086C_mRNA

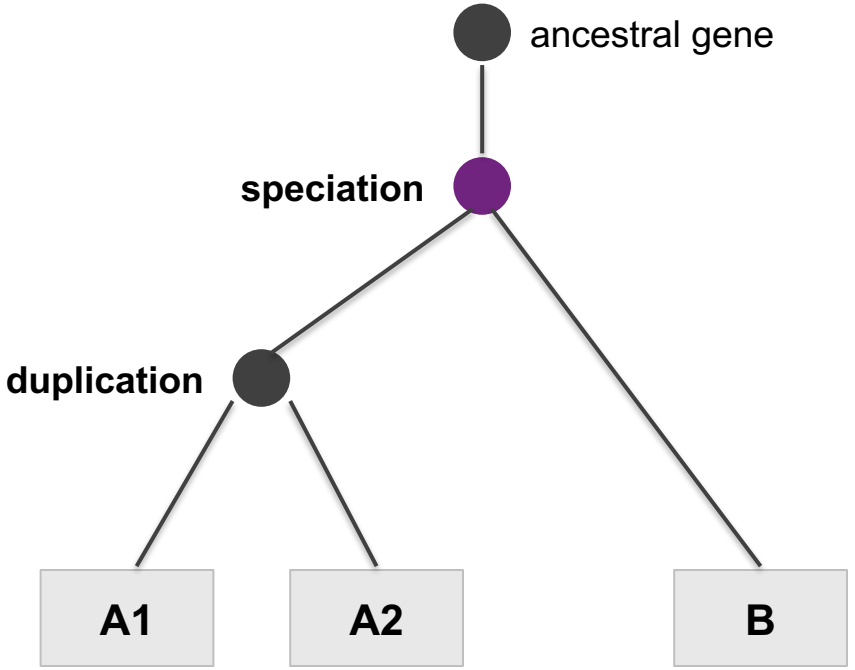
One feature per line; nine tab-separated columns

Parent= links exon and CDS to the mRNA, and the mRNA to the gene

A **gene model** = gene + mRNA + exons + CDS

Functional annotation: orthology

- Many predicted genes are "hypothetical proteins"
- **Homology**: shared ancestry; includes **paralogs**, which may have new functions
- **Orthology**: genes split by speciation usually keep their function
- eggNOG-mapper places each protein in an orthologous group and transfers annotations
- In: proteins (FASTA). Out: one table row per protein



A1 and B: orthologs (split by speciation)
A1 and A2: paralogs (split by duplication)

Reading the eggNOG table

Vocabulary	Describes	Example
GO	function, process, location	GO:0005355: glucose transmembrane transporter activity
KEGG	the gene family (KO) and its pathways	ADH1: KO K13953 in map00010 (Glycolysis / Gluconeogenesis)
EC	the reaction an enzyme catalyses	EC 1.1.1.1: alcohol dehydrogenase
COG	broad functional category (one letter)	K: transcription; P: inorganic ion transport

Example: ADH1 (S288C YOL086C): EC 1.1.1.1, KO K13953

Summary

- De novo assembly needs no reference; reference-guided is fast but biased
- Long reads span repeats; Flye assembles raw ONT reads; medaka polishes with the right model
- Judge an assembly by contiguity (N50), correctness (QUAST) and completeness (BUSCO)
- Lab 4 tells us how the wolf's genome scores on these three checks
- Structural annotation = where (GFF3); functional = what (eggNOG table)

Lab 4 preview: what you will do

1
Read the Flye log

2
Polish with medaka (sbatch)

3
Explore the supplied annotation

4
QUAST, then copy the report with scp

5
BUSCO and the legacy contrast

Bonus
Align the assembly to S288C; view in IGV

Questions are about reading results, not writing code

Coffee Break & Networking Session

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